



Society of Petroleum Engineers

SPE-229880-MS

Offshore Rigging Connection- Challenges and Solutions: A Case Study

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

Riggings are a set of items which serve as a connecting link between the structure that is lifted and the crane which lifts the structure. Conventionally the rigging is always connected to the structure onshore. However, in the current paper the rigging had to be connected offshore due to various reasons as outlined in the paper

Offshore rigging of heavy modules onto existing topsides presents formidable technical and operational challenges, especially when conventional rigging strategies cannot be applied. There can be many reasons such as space limitations, module configuration, centre of gravity (COG) offsets, or transport restrictions. This case study describes the engineering, execution, and lessons learned from a unique project that required rigging connection at offshore of a large module using synthetic slings with pre-installed shackles, facilitated by a custom-designed portal frame. By combining engineering design, planning, mechanical innovation, and safety-first execution, the team overcame conventional limitations and successfully completed the offshore lift. This paper offers a detailed examination of the problem, the engineering methods used, performance outcomes, safety enhancements, and recommendations for future similar offshore operations.

Introduction

Offshore brownfield installations demand lifting pre-assembled/ fabricated modules onto existing structures such as topsides etc. These modules can weigh several hundred tonnes, and the available interface points for lifting these modules—usually padeyes or trunnions—are limited in access and often not aligned with conventional rigging solutions to facilitate connection of Rigging at offshore.

These modules are lifted with a crane with the help of items called as "Rigging". The Riggings serve as a connection and creates a load path from the structure to the crane. Rigging is connected to strong points on the existing structure/module on one end and to the Crane hook at the other end. The strong points on the structure can be either a padeye or a trunnion depending on the project requirements.

Rigging generally has many components, such as slings, shackles, spreader bars, link plates etc. based on the module configuration and type of lift However majorly the components used in the project are briefly described below:

- **Sling,**
- **Shackles and**

- **Spreader bar.**

A **Sling** is a **flexible lifting device** used to **hold, support, and lift loads** during hoisting operations. It connects the **load to the lifting equipment** (such as a crane or hoist).

Common Types of Slings used in the industry are:

- **Wire rope sling** – made from steel wire ropes
- **Chain sling** – made of high-strength steel chain
- **Synthetic sling** – made from nylon or polyester webbing/round slings

A **shackle** is a **U-shaped metal connector** with a **removable pin or bolt** across the opening. It is used to **connect lifting slings, chains, or ropes to loads or lifting equipment**.

Common Types of shackles used in the industry are:

- **Anchor shackle (bow shackle)** – has a wider "O" shape, better for multi-directional loads
- **Chain shackle (D-shackle)** – narrower and used for straight-line pulls
- **Wide Body Shackle-** Have a **wider bow** providing efficient seating for slings.
- **Screw pin, bolt-type, or safety pin** varieties based on locking mechanism

A **spreader bar** is a **lifting device** used in conjunction with cranes to **lift heavy or large loads in a stable and balanced manner**. It is a **rigid horizontal beam** that spreads the load over multiple lifting points to reduce stress on the lifted object and prevent it from being crushed or deformed in addition to avoid clashed due to presence of existing equipment's on the module or the structure.

See below typical images of Slings and Shackles and Spreader bar.

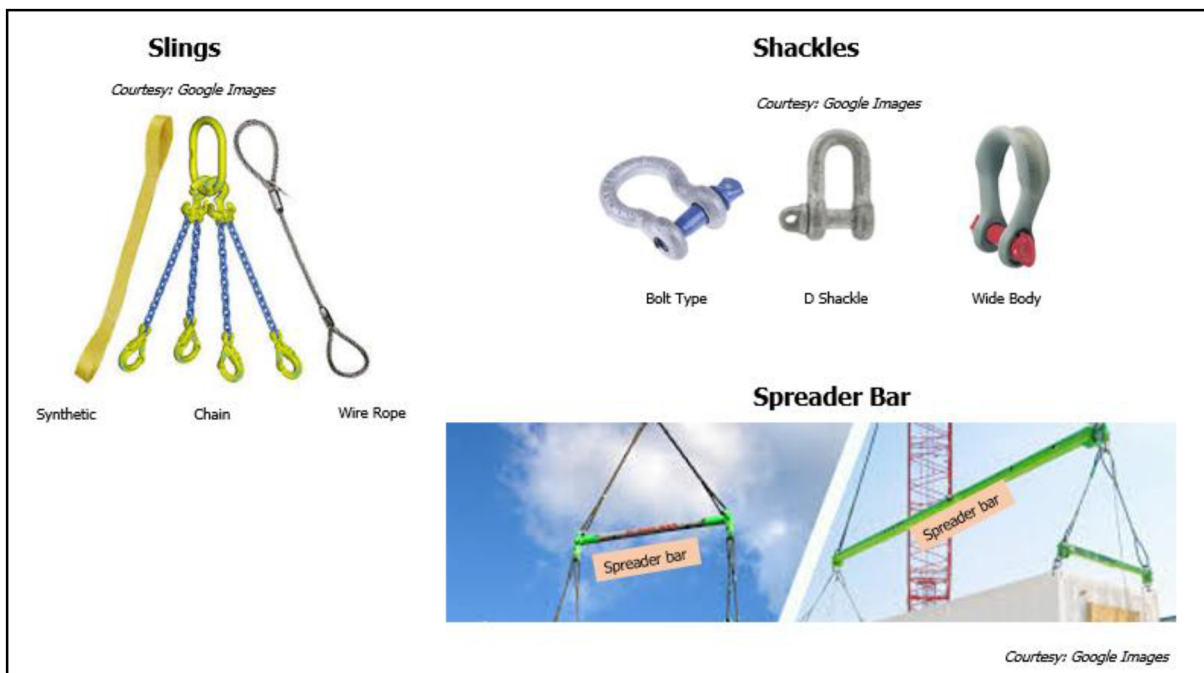


Figure 1—Typical types of Riggings

In conventional liftings these riggings are connected to the module in the fabrication yard and then transported offshore. At offshore the Heavy lift vessel (HLV) just connects the preinstalled Rigging on the module to the crane and finally installs the module/structure to its intended position.

In this specific project, the module's transport configuration, presence of major equipment's at top deck, limited support points for supporting the rigging elements, along with its COG and proximity to structural bracing, did not allow the rigging to be connected onshore in the fabrication yard. Offshore rigging, typically discouraged due to personnel safety risks and crane motion, became the only feasible path. This led to the development of a mechanical, rigging system using synthetic slings (instead of initially proposed wire Rope slings) and a rigid portal frame was used to ensure the rigging can be connected offshore. Thus creating a solution that fundamentally reimagined how offshore rigging installation can be approached in such scenarios.

Project Scope and Initial Constraints

Project Background

The project scope was to install 2 modules on a fixed existing platform by a heavy lift vessel (HLV). The module with fixed rigging weight amounted to around 600MT per module. The module lift was planned by using a spreader bar, which had 4 slings from the module connected to the bottom of Spreader bar and 2 slings from top of spreader bar connected to the crane.

The full rigging assembly comprised of 6 slings (4 at bottom of spreader bar and 2 slings at top of spreader bar), 10 heavy lift wide body shackles (4 on the module padeyes, 4 below the spreader bar padeyes and 2 above the spreader bar padeyes.)

See below Typical lifting arrangement for both the modules.

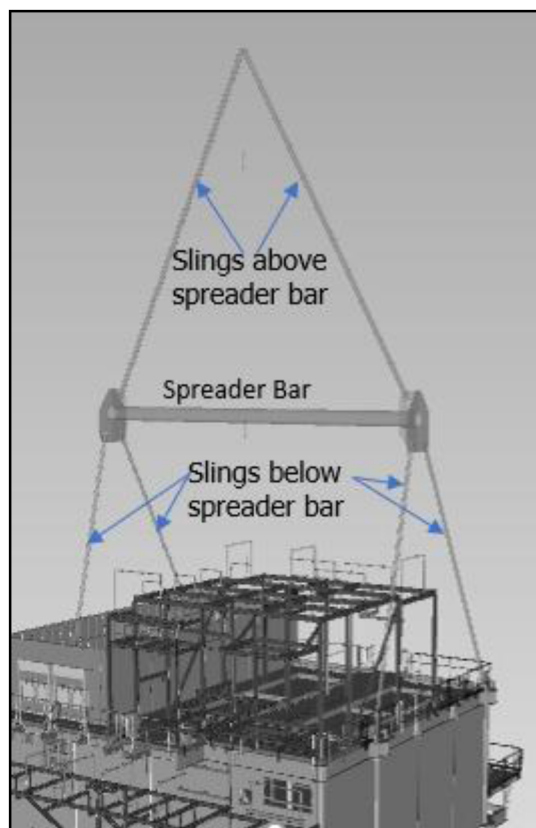


Figure 2—Typical Lifting Arrangement

The main challenge arose from the connection of the 4 shackles on the module padeyes with the full rigging assembly including slings and spreader bar hanging on the crane.

The client required a process module to be lifted onto a fixed offshore platform. The module, pre-assembled and weighing approximately 120 metric tonnes, featured four lifting padeyes. The challenge arose from the inability to pre-install rigging due to the module's frame interference and its transport cradle.

Why Rigging could not be connected Onshore?

To facilitate the rigging connection at the onshore location, it was initially considered to install the complete rigging arrangement on top of the existing module during transportation. However, the module houses multiple pieces of sensitive equipment along with high vertical ducts, which would have necessitated placing the rigging support structure at a considerable elevation above the top deck level. This elevated positioning would have significantly increased the overall centre of gravity (COG) of the transport assembly. Since the module's current configuration is highly sensitive to any rise in COG, especially during dynamic movements in transit, this posed a serious risk to transportation stability. After careful evaluation, it was concluded that placing any additional rigging support structure on the module itself was not a viable option, and alternative rigging or lifting strategies had to be considered to maintain safe transportation limits and also enable smooth connection of rigging at offshore.

Rigging Strategy Development

Based on the limitations of structure as stated above, rigging needed to be connected at offshore. Rigging arrangement was studied carefully to minimize the amount of rigging that can be connected offshore and maximize connection onshore.

On detailed evaluation it was decided that the 4 shackles will be preinstalled on the module and the balance rigging arrangement will be preassembled and transported to offshore on a cargo barge which will carry both the modules.

One of the main challenges was to connect the slings with the preinstalled shackles while the crane is rigged up with spreader bar and balance riggings.

Challenges and Constraints in connecting Rigging at offshore

Connecting the rigging offshore presented several operational and safety challenges that had to be carefully evaluated. One of the primary limitations was the initial plan to use wire rope steel slings, which are rigid, bulky, and extremely difficult to handle in offshore conditions. Manual lifting and positioning of these slings is also not feasible due to their substantial weight, making accurate alignment with pre-installed shackles highly challenging.

The situation was further complicated by the use of large heavy-duty shackles. Attempting to attach the shackles to the padeyes after being connected to the slings offshore was impractical, as the combined weight of the slings and shackles would've exceeded what could be safely or effectively managed by manual means.

Another significant constraint was the dynamic nature of offshore crane operations. Since the crane was mounted on a floating vessel, its hook block remains in constant motion due to wave-induced movements. This makes it extremely difficult to achieve precise manual alignment between the slings and shackles, increasing the risk of misalignment, equipment damage, or injury to personnel.

In addition to these challenges, the complex layout and geometry of the module itself posed further restrictions. The presence of pre-installed equipment and limited working access made it impossible to provide mechanical lifting guides or alignment aids that are typically used in such operations.

Collectively, these factors rendered the conventional approach to offshore rigging unsafe and operationally unfeasible. It became evident that significant design adaptations and procedural innovations were necessary to perform the rigging connection in a safe, efficient, and repeatable manner.

Strategic Approach to challenges

In response to the constraints and risks associated with offshore rigging, a targeted strategy was developed to prioritize both safety and operational efficiency. A key objective of the approach was to eliminate manual handling of heavy slings and shackles which is one of the hazardous aspects of traditional offshore lifting operations. To achieve this, the team opted for the use of lightweight, high-strength HMPE slings that could be easily maneuvered without heavy manual effort. This significantly reduces physical strain on personnel and allows for safer handling in confined offshore environments.

A central feature of the strategy was the implementation of a Portal frame over the module padeyes which can handle the shackles with the help of chain block thus avoiding manual handling of heavy shackles and allowing the sling eyes to be guided inside the preinstalled shackles. This not only minimizes the risk of misalignment and accidental slippage but also enables repeatable execution under dynamic offshore conditions.

To further enhance reliability, the entire rigging methodology was tested through mock-up trials during the yard phase. These trials were essential in validating the operational sequence, confirming equipment compatibility, and proactively identifying any safety or procedural concerns in a controlled setting. The insights gained from these tests were critical in refining the offshore execution plan and ensuring a smooth, incident-free lifting operation.

The following section outlines in detail the engineered solution and field implementation of this methodology.

Detailed Design and Execution Plan

Pre-Installed Shackles

The shackles were decided to be preinstalled in the module padeyes. This ensured that the shackles are also locked in position during transportation, thereby avoiding any additional sea fastening and extra handling of shackles offshore and avoiding manual handling of shackles at offshore.

Choice of Synthetic Slings

To overcome the handling and alignment challenges associated with traditional steel slings, synthetic slings made from High Modulus Polyethylene (HMPE) were selected for offshore rigging operations. These slings are ADNOC and DNV-GL type-approved and offer several distinct advantages over their steel counterparts. Most notably, HMPE slings are approximately 90% lighter than equivalent steel slings, significantly reducing the physical effort required for handling and positioning. This weight reduction enhanced both safety and efficiency during offshore operations. Additionally, their lightweight nature allowed for easier storage and transport, particularly for challenging offshore environments. The flexibility of HMPE slings also enabled better accommodation through clearances and complex geometries, which is critical when working around pre-installed equipment and tight rigging interfaces. These characteristics make synthetic slings a preferred choice for modern offshore lifting scenarios where safety, ease of use, and adaptability are essential.

Portal Frame with Chain Blocks

A custom-designed portal frame was utilized to facilitate safe and controlled offshore rigging over the module's lifting points. This structure was engineered to fit precisely above the designated padeyes, enabling accurate positioning without interference from surrounding equipment. The portal frame was equipped with chain blocks, which allowed for gradual and controlled lifting of the heavy-duty shackles, thus ensuring easy connection of slings inside the shackles. This mechanical advantage eliminated the need for additional lifting equipment like a crane etc and also provided a stable means of manoeuvring the shackle gradually above the padeyes to connect the sling eyes. Thus, by guiding the sling eye directly into the shackle bow

with precision, the system significantly reduced the risk of misalignment and enhanced the overall safety and repeatability of the rigging operation in dynamic offshore conditions.



Figure 3—TYP Portal frame.

Portal Frame with Chain Blocks-Challenges and Solutions

The module was equipped with four lifting padeyes, each located at a different corner and designated as P1, P2, P3, and P4. Each of these padeye locations presented unique spatial and structural challenges that required careful engineering and site adaptation for the installation of portal frames with chain blocks.

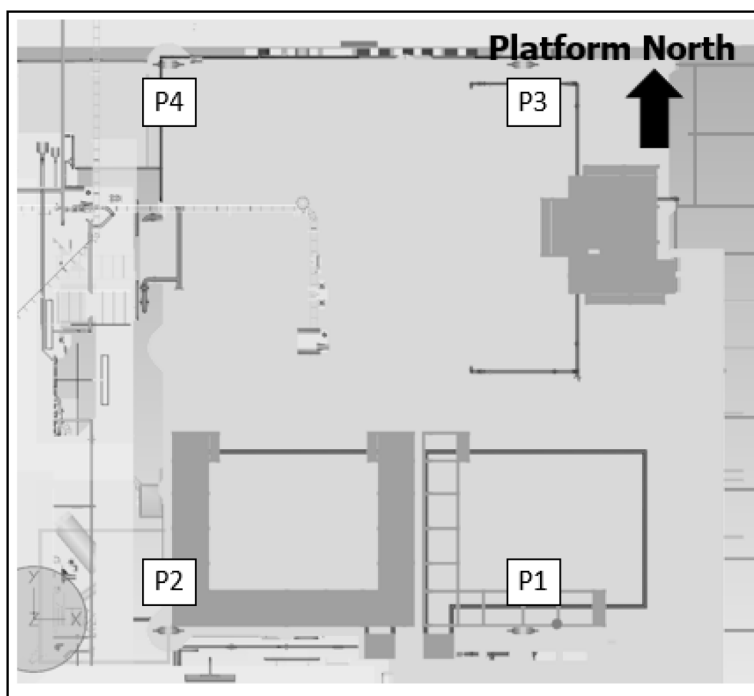


Figure 4—Padeye Numbering

P1 Padeye: South East Padeye

The **P1 padeye**, located at the southeast corner of the module, posed access constraints due to the congestion of surrounding equipment and supports. This padeye was situated close to internal equipment located

within the module. The top deck area immediately above the padeye was covered with thermal insulation underneath the top deck, making it **impractical to perform any welding on top of the module**.

To overcome this, a suitable mounting point for the portal frame was selected using the **existing equipment support structure**. This approach allowed for mechanical load transfer without disturbing the insulated surfaces. The portal frame was designed to span the area efficiently while avoiding interference with the adjacent installations.

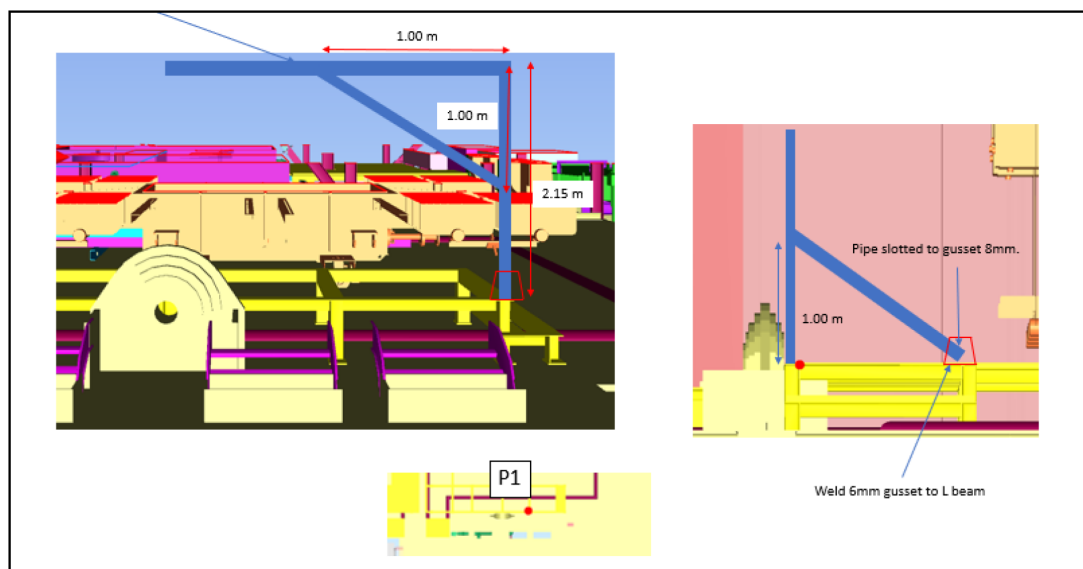


Figure 5—Typical Portal frame at P1 Padeye.

P2 Padeye: Southwest Padeye

The **P2 padeye**, positioned at the southwest edge of the module, had its own set of spatial constraints. It was located near a corner where two adjacent sides were occupied — two sides by an **existing platform**. The third side was aligned in the direction of the sling, and the fourth side was partially obstructed by existing **firewall /heavy equipment** which was the only feasible location to provide the portal frame

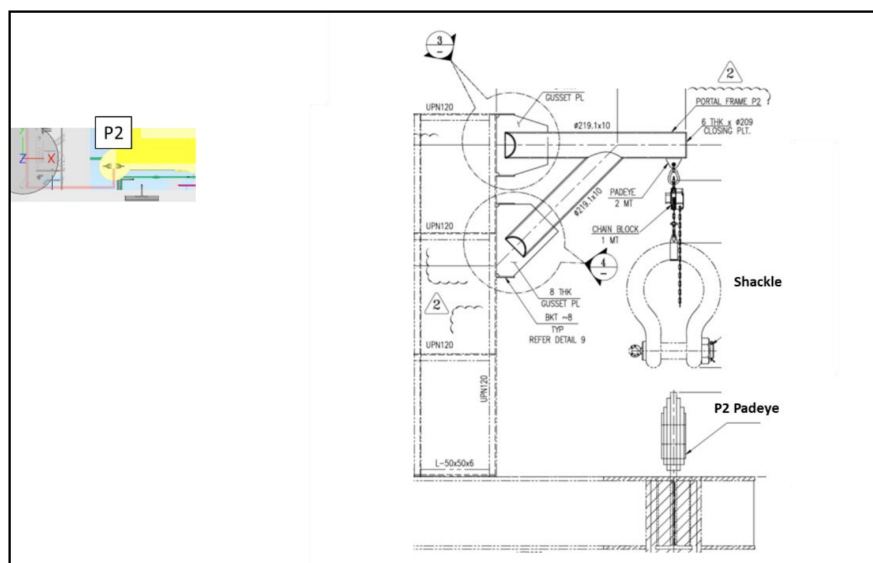


Figure 6—Typical Portal frame at P2 Padeye.

With these limitations, the portal frame was designed to integrate with the **existing firewall structure** on one side. However, since this integration introduced lateral load paths through a non-structural firewall, **additional reinforcement was required** to ensure effective and safe load transfer. This included stiffening measures to avoid local failure of the firewall interface.

P3 Padeye: Northeast padeye

The **P3 padeye**, was located on the Northeast side of the module. One side faced the open sea, the second was obstructed by equipment, and the third was occupied by the sling orientation. The **fourth side had an existing equipment support**, which was considered for portal placement.

At P3 the portal frame was extended directly from the existing equipment support. This integration was possible because the geometry and accessibility of the support allowed a more seamless structural link. The portal was aligned and fixed to utilize the support's capacity without additional welding interfaces, simplifying the installation.

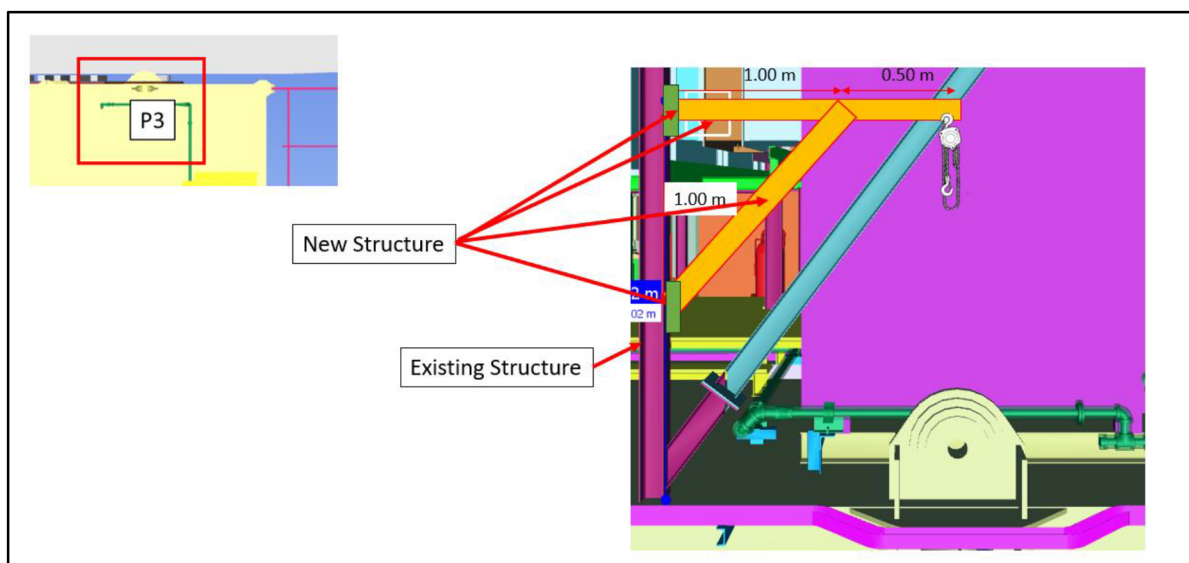


Figure 7—Typical Portal frame at P3 Padeye

P4 Padeye: Northwest padeye

Located at the northeast edge, the **P4 padeye** was bounded on two sides by open sea, and on a third side by the sling direction. The only usable side was the fourth side, where an **existing equipment support** was present.

Given the lack of structural access on three sides, the portal was **strategically placed** over this equipment support. However, the support comprised a **C-channel section**, making direct load placement over the flange unfeasible. To solve this, a **gusset plate arrangement** was introduced. The vertical portal member was connected to the gusset, which was then **welded onto the web of the channel**, ensuring that the flange wasn't overstressed and that the load was properly transferred without compromising the integrity of the equipment support.

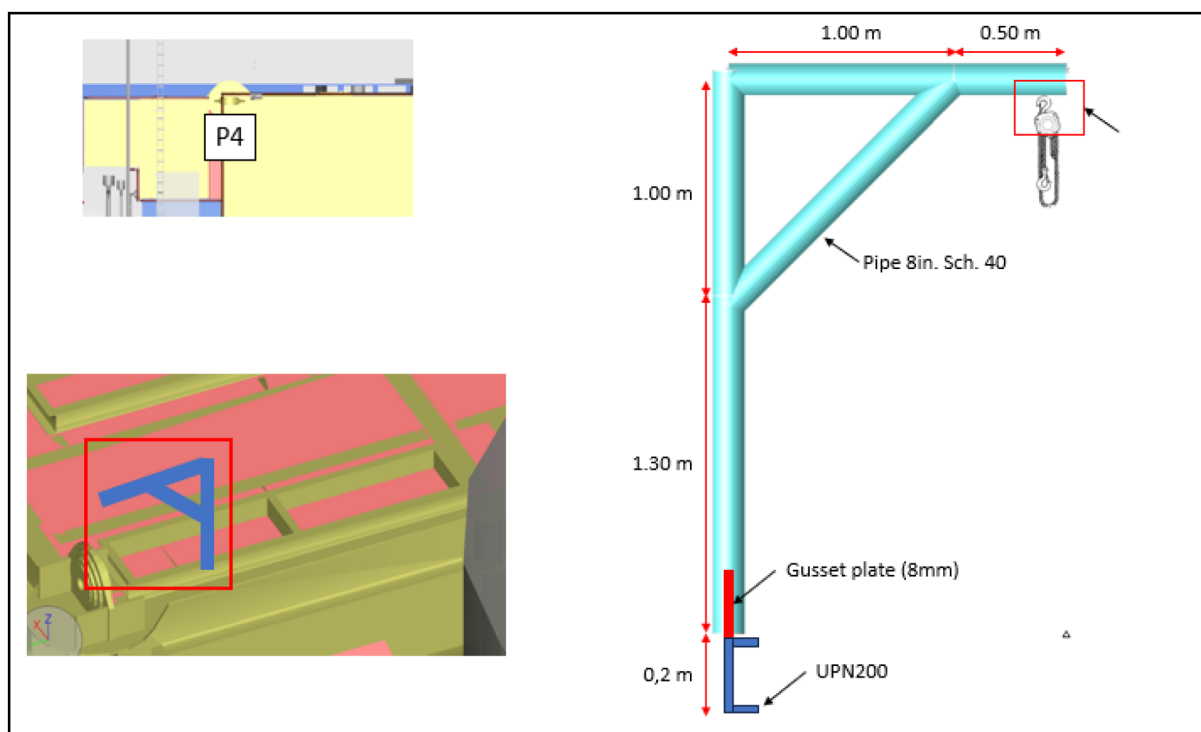


Figure 8—Typical Portal frame at P4 Padeye.

Each of the four padeye locations presented unique installation challenges driven by equipment layout, insulation restrictions, structural accessibility, and edge positioning. The portal frames were custom-engineered at each location to leverage existing supports while ensuring safe and effective sling alignment and load transfer using chain blocks.

Results and Operational Observations

The use of portal frames with chain blocks led to significant improvements in safety, efficiency, and operational control during the offshore rigging and lifting operations.

Improved Safety

The need for riggers to manually insert slings under a live crane hook is eliminated, thereby reducing direct exposure to suspended loads. Dynamic risks arising from crane movement in offshore conditions were mitigated through the use of chain blocks, which enabled precise and controlled alignment. Additionally, the adoption of lightweight synthetic slings improved handling ergonomics and reduced the likelihood of strain or injury.

Faster Execution

Much of the rigging was pre-installed during yard preparation, significantly minimizing adjustments required offshore. The inclusion of working platforms and chain blocks allowed all padeye connections to be made simultaneously in a controlled manner. This not only streamlined the sequence of operations but also leads to a reduction in overall rigging and lifting time.

Enhanced Operational Control

The portal frame design enabled fine-tuning through chain blocks, giving riggers full control over sling insertion without relying on crane swings. This method introduced repeatability and reduced variability, allowing the entire rigging process to be executed with higher confidence and precision.

Use of Same set of Rigging for both modules

The use of portal frames enabled the reuse of the same set of rigging and spreader bar for both modules (excluding the four shackles fixed on the module padeyes). This approach eliminated the need to procure additional rigging sets or fabricate a second spreader bar, resulting in significant cost savings, reduced fabrication efforts, and time efficiency during offshore operations. By optimizing resource usage, the team avoids delays in offshore rigging connection thereby streamlining the entire lifting process with minimal duplication of equipment.

Lessons Learned

The offshore lifting operation provides a valuable insight that can be applied to future projects to enhance safety, efficiency, and precision.

One of the key challenges was the use of traditional **steel slings**, which proved to be excessively heavy and difficult to handle in the offshore environment. By switching to **synthetic slings**, the team significantly reduced manual handling effort and improved overall rigging ergonomics.

Another critical issue was the **difficulty in aligning slings at the padeyes** due to crane motion and module movement. This is effectively resolved by employing a **stable portal frame integrated with chain blocks**, allowing for fine adjustments and safe, controlled alignment at each lifting point.

Safety risks are also significantly mitigated. Traditionally, riggers can be exposed to suspended loads while inserting slings manually if rigging is not pre-installed. The introduction of **mechanized sling insertion using chain blocks** eliminates the need for personnel to operate directly beneath the crane hook, thereby enhancing safety.

Lastly, **offshore delays caused by rigging rework** were addressed by **pre-installing shackles and customizing handling arrangements** during yard preparation. This proactive approach enabled faster execution offshore, minimized downtime, and established a more streamlined, repeatable rigging procedure.

Implications for Future Projects

This case study presents a **scalable and replicable solution** for offshore rigging in scenarios where conventional pre-rigging is not practical due to space limitations, insulation constraints, or dynamic environmental conditions. The successful execution of the operation was made possible through a **combination of early planning, innovation, and procedural discipline**.

Key success factors included the **early identification of offshore rigging challenges** during the detailed design stage, allowing sufficient time for engineering adaptations. Proactive measures taken in the **fabrication yard**, such as the **pre-installation of shackles** and padeye access modifications, played a crucial role in minimizing offshore interventions.

The use of **lightweight synthetic slings** and **custom-designed portal frames** with chain blocks enhanced both ergonomics and operational safety, making the system easier to handle and reducing offshore exposure.

Equally important was the focus on **rigging methodology development**, including **structured planning, sequence reviews, and personnel training**, which helped standardize the operation and minimize execution variability.

Overall, this case provides a **template for future offshore lifting projects**, especially for congested or constrained module layouts, where **safety, efficiency, and repeatability** are key performance drivers. Adopting this method can lead to substantial improvements in offshore execution timelines and cost optimization.

Conclusion

The adopted methodology successfully addressed the challenges of offshore rigging under complex and constrained conditions. By integrating lightweight synthetic slings, a custom-designed portal frame with mechanical insertion systems, and pre-rigging efforts at the fabrication yard, the module lift was executed safely and efficiently. This approach significantly reduced manual intervention and offshore alignment risks, while enhancing control and repeatability. The overall strategy serves as a valuable reference model for future offshore projects where rigging logistics are limited by equipment geometry, access restrictions, or motion-induced hazards, offering a scalable and practical solution for similar lifting operations.

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